Analysis III Übungsblatt 5 Solutions

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> *Uebungsblatt 05 - PDF*

Solve all exercises of the loaded exercise sheet inlcuding recap of each actual task and a step by step derivation. The output should be in correct LaTeX syntax using the preamble given below. Put your output into a code block separated from your comments.

--- minimlal header for subfiles format--

\documentclass[../akolberesultsmaster.tex]{subfiles}

\begin{document}

> *Thinking:*
>
> > *Thinking:*
> >
> > ***Reviewing The Input***
> >
> > *I'm currently focused on the initial analysis of the PDF document. I'm carefully examining the structure of the input file and identifying the specific tasks outlined within it. My primary objective is to gain a clear understanding of the exercises. This stage is crucial for developing a complete approach to solving the presence and homework tasks.*
> >
> > ***Examining the Details***
> >
> > *I'm now fully immersed in the exercise details. I've pinpointed the LaTeX format with its specific subfile structure. The German language is confirmed as the appropriate choice for the solution. I'm focusing on "Exercise Sheet 5" from Uni Köln's Analysis III course, led by Prof. Sabatini, with a deadline of*

November 24, 2025. Specifically, I'm concentrating on Präsenzaufgabe 1, which involves extending a Lebesgue-integrable function.

Investigating Linearity & Integrability

I'm now deeply engaged with Präsenzaufgabe 2 (P2) and Hausaufgabe 1 (H1). P2's linearity property is straightforward. H1's focus on integrability of f^+ and $|f|$ is interesting. The recent OCR typos in Hausaufgabe 3 (H3) are clarified with standard measure theory context. I've rephrased this section more rigorously. I'm starting to analyze the alternating harmonic series-based function in H4.

Addressing Absolute Convergence

I'm now focusing on Hausaufgabe 4 (H4). I've confirmed that the function's construction relates to the alternating harmonic series. I'm carefully examining the Riemann integrability on the interval $[0, 1]$ and proving that this is not Lebesgue integrable, considering absolute convergence requirements. My strategy involves analyzing the function's structure.

Considering Integrability Definitions

I'm now revisiting the definition of Lebesgue integrability, specifically focusing on how the absolute value is addressed. I'm considering approaches using either the Monotone or Dominated Convergence Theorems, or a more basic approach via simple functions. I've realized that the reference to "Lemma 26" suggests a standard proof using simple functions might be appropriate. I'm solidifying my approach to H1, ensuring I adhere to the common definition of integrability, especially where absolute value is involved. I will explore a proof assuming the standard construction via simple functions sequence.

Examining Integrability Proofs

I'm now focusing on solidifying the integrability proofs for Hausaufgabe 1 (H1) and Hausaufgabe 2 (H2). For H1, I've confirmed that the definitions of f, f^+ and f^- apply. I'm exploring the connection between Riemann and Lebesgue integrability in H2, leveraging Lebesgue's criterion. I'm focusing on

> *the boundedness aspect and its implications for domination by a constant function. For H3, I've worked through the measurability arguments for both parts, considering generators and function composition. For the final exercise H4, I'm working to produce a counterexample.*
>
> ***Developing a Counterexample***
>
> *I'm currently working to prove that the function from H4 is not Lebesgue integrable, given that it is Riemann integrable. Focusing on the intervals* $I_n = (1/(n+1), 1/n]$ *, I've now defined the function* $f(x) = (-1)^n / x$ *on these intervals. I've shown the integral on each interval is* $(-1)^n \ln(1+1/n)$ *, converging via the alternating series test. However, the absolute integral diverges as the harmonic series. I need to complete this proof with proper mathematical notation.*
>
> ***Analyzing Set-Theoretic Notation***
>
> *I'm now fully immersed in the notation used throughout the problem set. I've successfully clarified the meaning of "* $A - B(R)$ *" to represent the standard measurable function context. This is crucial for constructing accurate and rigorous solutions. I will diligently pay close attention to other math symbols throughout the remaining tasks. My current focus is refining the solution approach for each exercise, building upon my work thus far.*
>
> ***Refining Simple Function Approach***
>
> *I'm now refining the simple function approach for the Lebesgue integrability proof in H4. I've defined the intervals* J_k *and established bounds for* x *and* $1/\sqrt{x}$ *on these intervals. I'm constructing simple functions based on these bounds. The geometric series will likely play a key role in the final calculation of the Lebesgue integral. I'm meticulously checking the details to ensure the argument is sound and rigorously justified.*

Here are the solutions to the exercises from "Übungsblatt 05" for Analysis III. The solutions are formatted in LaTeX as requested, including the task recap and step-by-step derivations.

Analysis and Approach

1. **Präsenzaufgabe 1:** Focuses on the definition of simple functions. The extension is trivial by setting the function to zero outside the original domain X , as the integral of zero is zero.
2. **Präsenzaufgabe 2 (Linearity):** This is a standard property of the Lebesgue integral. The proof typically proceeds from simple functions (where linearity is immediate from the definition) to non-negative functions (using the Monotone Convergence Theorem) and finally to general integrable functions by splitting into positive and negative parts.
3. **Hausaufgabe 1 (Absolute Value):** Relies on the decomposition $f = f^+ - f^-$. Since integrability usually requires $\int f^+ < \infty$ and $\int f^- < \infty$, the sum (which is $|f|$) is also integrable.
4. **Hausaufgabe 2 (Riemann vs. Lebesgue):** A classic theorem. Since the domain is bounded, a Riemann-integrable function is bounded. Step functions can approximate it from above and below. The key is that Riemann integrability implies continuity almost everywhere, or simply using the definition of Upper/Lower sums as integrals of simple functions that converge to the same value.
5. **Hausaufgabe 3 (Measurability):**
   - Part (a) is a generator argument. (a, ∞) generates the Borel σ -algebra.
   - Part (b) uses the property that the composition of measurable functions (specifically with a continuous function $h(y) = 1/y$ on R\setminus \left{0\right}) is measurable.
6. **Hausaufgabe 4 (Counterexample):** A variation of the alternating harmonic series.
   - (a) It converges as an improper Riemann integral because the alternating series $\sum(-1)^n/n$ converges.
   - (b) It is not Lebesgue integrable because Lebesgue requires absolute convergence, and $\sum 1/n$ diverges.
7. **Hausaufgabe 5 (Approximation):** Explicit construction of simple functions to integrate $x^{-1/2}$. We will use the geometric partition hinted at in the text to

set up geometric series for the upper and lower sums.

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\documentclass[../akolberesultsmaster.tex]{subfiles}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}

\begin{document}

\section*{Analysis III - Übungsblatt 5}

\subsection*{Präsenzaufgabe 1}
\textbf{Aufgabe:}
Sei $g: X \rightarrow \mathbb{R}$ eine einfache Lebesgue-
integrierbare Funktion. Zeigen Sie, dass für jedes n-Intervall $Q
\subseteq \mathbb{R}^n$ mit $X \subseteq Q$ eine einfache
Lebesgue-integrierbare Funktion $\tilde{g}: Q \rightarrow
\mathbb{R}$ existiert, sodass $\tilde{g}|_X = g$ und
$$
\int_X g ~d\lambda = \int_Q \tilde{g} ~d\lambda.
$$

\textbf{Lösung:}
Da $g: X \rightarrow \mathbb{R}$ eine einfache Funktion ist, hat
sie die Form:
$$
g(x) = \sum_{i=1}^k c_i \chi_{A_i}(x)
$$
wobei $c_i \in \mathbb{R}$ und $A_i \subseteq X$ messbare Mengen
sind. Da $g$ Lebesgue-integrierbar ist, gilt $\lambda(A_i) <
\infty$ für alle $i$ mit $c_i \neq 0$.

Wir definieren nun die Erweiterung $\tilde{g}: Q \rightarrow
\mathbb{R}$ durch:
$$
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\tilde{g}(x) := \begin{cases}
g(x) & \text{falls } x \in X \\
0 & \text{falls } x \in Q \setminus X
\end{cases}
$$
Da $X \subseteq Q$, ist diese Funktion wohldefiniert.
Wir können $\tilde{g}$ ebenfalls als Linearkombination
charakteristischer Funktionen schreiben. Die Mengen $A_i$ sind
Teilmengen von $X$ und damit auch Teilmengen von $Q$.
$$
\tilde{g}(x) = \sum_{i=1}^k c_i \chi_{A_i}(x) + 0 \cdot \chi_{Q
\setminus X}(x) = \sum_{i=1}^k c_i \chi_{A_i}(x).
$$
(Hierbei betrachten wir die $A_i$ nun als Teilmengen des Raumes
$Q$).
Da die Struktur der Summe identisch bleibt und die Maße
$\lambda(A_i)$ unabhängig davon sind, ob wir $A_i$ als Teilmenge
von $X$ oder $Q$ betrachten (das Lebesgue-Maß ist auf
$\mathbb{R}^n$ definiert), ist $\tilde{g}$ einfach und messbar.

Für das Integral gilt:
$$
\int_Q \tilde{g} ~d\lambda = \sum_{i=1}^k c_i \lambda(A_i) =
\int_X g ~d\lambda.
$$
Somit ist die Aussage gezeigt.

\hrule

\subsection*{Präsenzaufgabe 2 (Lemma 26)}
\textbf{Aufgabe:}
Seien $f, g: X \rightarrow \mathbb{R}$ Lebesgue-integrierbare
Funktionen. Zeigen Sie, dass dann auch $\alpha f + \beta g$
Lebesgue-integrierbar ist für alle $\alpha, \beta \in \mathbb{R}$
und gilt:
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$$
\int_X (\alpha f + \beta g) ~d\lambda = \alpha \int_X f ~d\lambda
+ \beta \int_X g ~d\lambda.
$$

\textbf{Lösung:}
Wir beweisen dies in zwei Schritten: Homogenität und Additivität.

1. \textbf{Homogenität:} Sei $f$ integrierbar und $\alpha \in
\mathbb{R}$.
   Ist $\alpha = 0$, so ist $\alpha f = 0$, was integrierbar ist
mit Integral 0.
   Ist $\alpha \neq 0$, betrachten wir zuerst einfache Funktionen
$\phi_n$. Sei $(\phi_n)_{n \in \mathbb{N}}$ eine Folge einfacher
Funktionen mit $\phi_n \rightarrow f$ und $\int |\phi_n - f|
d\lambda \rightarrow 0$.
   Dann konvergiert die Folge einfacher Funktionen $(\alpha
\phi_n)_n$ gegen $\alpha f$.
   Für einfache Funktionen ist die Linearität klar: $\int \alpha
\phi d\lambda = \alpha \int \phi d\lambda$.
   Durch den Grenzübergang (unter Nutzung der
Stetigkeitseigenschaften des Integrals bzw. Definition über $L^1$-
Cauchy-Folgen) folgt:
   $$ \int (\alpha f) d\lambda = \alpha \int f d\lambda. $$

2. \textbf{Additivität:} Seien $f, g$ integrierbar.
   Nach Definition ist $f$ integrierbar, wenn $f = f^+ - f^-$ mit
$\int f^+ < \infty$ und $\int f^- < \infty$.
   Zunächst zeigen wir es für nicht-negative Funktionen $f, g \geq
0$.
   Es existieren monoton wachsende Folgen einfacher Funktionen
$(\phi_n)$ und $(\psi_n)$ mit $\phi_n \nearrow f$ und $\psi_n
\nearrow g$.
   Dann ist $(\phi_n + \psi_n)$ eine Folge einfacher Funktionen,
die monoton gegen $f+g$ konvergiert.
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   Nach dem Satz von der monotonen Konvergenz (B. Levi) gilt:
   $$
   \int (f+g) d\lambda = \lim_{n \to \infty} \int (\phi_n + \psi_n) d\lambda.
   $$
   Da die Linearität für einfache Funktionen gilt:
   $$
   = \lim_{n \to \infty} \left( \int \phi_n d\lambda + \int \psi_n d\lambda \right) = \int f d\lambda + \int g d\lambda.
   $$
   Für allgemeine $f, g$ zerlegen wir in Positiv- und Negativteile oder nutzen die Definition der Integrierbarkeit über $L^1(\lambda)$ als Vektorraum, wobei die Dreiecksungleichung $|(f+g)| \leq |f| + |g|$ die Integrierbarkeit der Summe sichert.

   Zusammenfassend folgt für beliebige $\alpha, \beta$:
   $$ \int (\alpha f + \beta g) d\lambda = \int \alpha f d\lambda + \int \beta g d\lambda = \alpha \int f d\lambda + \beta \int g d\lambda. $$

\hrule

\subsection*{Hausaufgabe 1 (7 Punkte)}
\textbf{Aufgabe:}
Sei $f: X \rightarrow \mathbb{R}$ Lebesgue-integrierbar. Zeigen Sie, dass dann auch $f^+$ und $|f|$ Lebesgue-integrierbar sind.

\textbf{Lösung:}
Die Definition der Lebesgue-Integrierbarkeit für eine messbare Funktion $f$ besagt üblicherweise, dass das Integral des Betrags endlich ist, also $\int_X |f| d\lambda < \infty$, oder äquivalent, dass sowohl der Positivteil $f^+$ als auch der Negativteil $f^-$ ein endliches Integral besitzen.

Gehen wir davon aus, dass die Definition der Integrierbarkeit ist:
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$f$ ist integrierbar, wenn $\int f^+ d\lambda < \infty$ und $\int
f^- d\lambda < \infty$.
In diesem Fall ist die Aussage trivial, da $f^+$ per Definition
integrierbar ist.
Für $|f|$ gilt: $|f| = f^+ + f^-$.
Da $f^+$ und $f^-$ integrierbar sind, ist nach der Linearität des
Integrals (siehe Präsenzaufgabe 2) auch deren Summe $|f|$
integrierbar.

Alternativ, falls die Definition über den Abschluss einfacher
Funktionen erfolgt:
Da $f$ messbar ist, sind auch $f^+ = \max(0, f)$ und $|f|$ messbar
(Verkettung messbarer Funktionen mit stetigen Funktionen).
Es gilt punktweise $0 \leq f^+(x) \leq |f(x)|$.
Da $f$ integrierbar ist, existiert $\int f d\lambda$. Die
Integrierbarkeit impliziert in der Regel $\int |f| d\lambda <
\infty$.
Ist $f$ integrierbar, so ist $f \in \mathcal{L}^1$. Das ist
äquivalent zu $|f| \in \mathcal{L}^1$.
Da $0 \le f^+ \le |f|$, folgt aus der Integrierbarkeit von $|f|$
(Majorantenkriterium) direkt die Integrierbarkeit von $f^+$.

Zusammenfassend:
1. $f$ integrierbar $\implies f$ messbar.
2. $|f|(\cdot) = |f(\cdot)|$ ist messbar und $f^+(\cdot)$ ist
messbar.
3. Es gilt $|f| = f^+ + f^-$. Da $f$ integrierbar ist, sind $\int
f^+ < \infty$ und $\int f^- < \infty$.
4. Daraus folgt $\int |f| = \int f^+ + \int f^- < \infty$. Also
ist $|f|$ integrierbar.
5. Da $\int f^+ < \infty$, ist $f^+$ integrierbar.

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\subsection*{Hausaufgabe 2 (Lemma 27) (10 Punkte)}
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\textbf{Aufgabe:}
Sei $X \subseteq \mathbb{R}^n$ beschränkt und $f: X \rightarrow \mathbb{R}$ eine Riemann-integrierbare Funktion. Zeigen Sie, dass dann $f$ Lebesgue-integrierbar ist und gilt:
$$
\int_X f ~d\lambda = \int_X f(\underline{x}) d\underline{x} \quad (\text{Riemann-Integral}).
$$

\textbf{Lösung:}
Da $f$ auf dem beschränkten Gebiet $X$ (bzw. einem Quader $Q \supseteq X$, wobei $f$ außerhalb von $X$ als 0 fortgesetzt wird) Riemann-integrierbar ist, ist $f$ beschränkt. Sei $|f(x)| \leq M$.
Da $X$ beschränkt ist, hat $X$ endliches Maß $\lambda(X) < \infty$.

1. **Lebesgue-Integrierbarkeit:**
   Eine Funktion ist genau dann Riemann-integrierbar, wenn sie fast überall stetig ist (Lebesguesches Kriterium). Da sie fast überall stetig ist, ist sie $\lambda$-fast überall gleich einer messbaren Funktion (oder direkt Borel-messbar bis auf eine Nullmenge), also Lebesgue-messbar.
   Da $f$ beschränkt ist ($|f| \leq M$) und auf einer Menge endlichen Maßes definiert ist, ist die konstante Funktion $g(x) = M$ eine integrierbare Majorante:
   $$ \int_X |f| d\lambda \leq \int_X M d\lambda = M \cdot \lambda(X) < \infty. $$
   Folglich ist $f$ Lebesgue-integrierbar.

2. **Gleichheit der Integrale:**
   Sei $f$ R-integrierbar auf einem Quader $Q$ (durch triviale Fortsetzung).
   Es existiert eine Folge von Treppenfunktionen (Step-functions) $\phi_k$ und $\psi_k$ mit $\phi_k \leq f \leq \psi_k$, sodass für die Riemann-Integrale gilt:
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```latex
   $$ \lim_{k \to \infty} \int_Q \phi_k(x) dx = \lim_{k \to \infty} \int_Q \psi_k(x) dx = \int_Q^{R} f(x) dx. $$
   Treppenfunktionen sind einfache Funktionen. Für diese gilt, dass das Riemann-Integral gleich dem Lebesgue-Integral ist: $\int \phi_k dx = \int \phi_k d\lambda$.
   Aus $\phi_k \leq f \leq \psi_k$ folgt durch die Monotonie des Lebesgue-Integrals:
   $$ \int \phi_k d\lambda \leq \int f d\lambda \leq \int \psi_k d\lambda. $$
   Da die linke und rechte Seite gegen das Riemann-Integral konvergieren, muss nach dem Einschnürungssatz gelten:
   $$ \int_X f d\lambda = \int_X f(x) dx. $$

\hrule

\subsection*{Hausaufgabe 3 (9 Punkte)}
\textbf{Aufgabe:}
Sei $(X, \mathcal{A})$ ein messbarer Raum.
a) Zeigen Sie, dass eine Funktion $f: (X, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B})$ genau dann $\mathcal{A}-\mathcal{B}$-messbar ist, wenn $f^{-1}((a, +\infty)) \in \mathcal{A}$ für alle $a \in \mathbb{R}$.
b) Sei $g: (X, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B})$ eine $\mathcal{A}-\mathcal{B}$-messbare Funktion. Zeigen Sie, dass ebenfalls $\frac{1}{g}$ $\mathcal{A}-\mathcal{B}$-messbar ist, falls $g(x) \neq 0$ für alle $x \in X$.

\textbf{Lösung:}

\textbf{a)}
Die Borel-$\sigma$-Algebra $\mathcal{B}(\mathbb{R})$ wird von den offenen Intervallen der Form $(a, \infty)$ erzeugt. Das heißt $\mathcal{B} = \sigma(\mathcal{E})$ mit dem Erzeugendensystem $\mathcal{E} = \{ (a, \infty) \mid a \in \mathbb{R} \}$.
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"$\Rightarrow$": Sei $f$ messbar. Nach Definition gilt $f^{-1}(B)
\in \mathcal{A}$ für alle $B \in \mathcal{B}$. Da $(a, \infty) \in
\mathcal{B}$, ist $f^{-1}((a, \infty)) \in \mathcal{A}$.

"$\Leftarrow$": Sei $f^{-1}((a, \infty)) \in \mathcal{A}$ für alle
$a \in \mathbb{R}$.
Wir betrachten das Mengensystem $\mathcal{C} = \{ B \subseteq
\mathbb{R} \mid f^{-1}(B) \in \mathcal{A} \}$.
Man prüft leicht nach, dass $\mathcal{C}$ eine $\sigma$-Algebra
ist (Urbilder vertauschen mit Komplementen und Vereinigungen).
Da nach Voraussetzung das Erzeugendensystem $\mathcal{E} \subseteq
\mathcal{C}$ ist, muss auch die davon erzeugte $\sigma$-Algebra
enthalten sein: $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{E})
\subseteq \mathcal{C}$.
Das bedeutet, für alle $B \in \mathcal{B}$ ist $f^{-1}(B) \in
\mathcal{A}$. Also ist $f$ messbar.

\textbf{b)}
Sei $g: X \rightarrow \mathbb{R}$ messbar und nullstellenfrei.
Wir definieren die Funktion $h: \mathbb{R} \setminus \{0\}
\rightarrow \mathbb{R}$ durch $h(y) = \frac{1}{y}$.
Die Funktion $h$ ist auf ihrem Definitionsbereich stetig. Stetige
Funktionen sind Borel-messbar.
Die Funktion $\frac{1}{g}$ ist die Komposition $h \circ g$.
Da die Verkettung einer messbaren Funktion ($g$) mit einer
messbaren Funktion ($h$) wieder messbar ist, ist $\frac{1}{g}$
messbar.
(Formal: Sei $B \in \mathcal{B}$. Dann ist $(h \circ g)^{-1}(B) =
g^{-1}(h^{-1}(B))$. Da $h$ messbar, ist $h^{-1}(B) \in
\mathcal{B}$. Da $g$ messbar, ist das Urbild dieser Menge in
$\mathcal{A}$).

\hrule

\subsection*{Hausaufgabe 4 (11 Punkte)}
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\textbf{Aufgabe:}
Sei $f: (0, 1] \rightarrow \mathbb{R}$ definiert durch
$$ f(x) := (-1)^n \cdot \frac{1}{x} \quad \text{für } x \in
\left(\frac{1}{n+1}, \frac{1}{n}\right], n \in \mathbb{N}. $$

\textbf{a) Existenz des uneigentlichen Riemann-Integrals:}
Das Integral ist definiert als $\lim_{A \to \infty} \sum_{n=1}^A
\int_{1/(n+1)}^{1/n} f(x) dx$.
Berechnen wir das Integral auf einem Teilintervall $I_n =
(\frac{1}{n+1}, \frac{1}{n}]$:
$$ \int_{1/(n+1)}^{1/n} (-1)^n \frac{1}{x} dx = (-1)^n \left[
\ln(x) \right]_{1/(n+1)}^{1/n} = (-1)^n \left( \ln\left(\frac{1}
{n}\right) - \ln\left(\frac{1}{n+1}\right) \right) $$
$$ = (-1)^n \ln\left( \frac{1/n}{1/(n+1)} \right) = (-1)^n
\ln\left( \frac{n+1}{n} \right) = (-1)^n \ln\left( 1 + \frac{1}{n}
\right). $$
Das uneigentliche Integral ist somit die Reihe:
$$ \sum_{n=1}^{\infty} (-1)^n \ln\left( 1 + \frac{1}{n} \right).
$$
Da $\ln(1 + \frac{1}{n})$ eine monoton fallende Nullfolge ist (für
$n \to \infty$ geht der Term gegen $\ln(1)=0$), konvergiert diese
alternierende Reihe nach dem Leibniz-Kriterium.
Das uneigentliche Riemann-Integral existiert also.

\textbf{b) Keine Lebesgue-Integrierbarkeit:}
Damit $f$ Lebesgue-integrierbar ist, muss $\int_{(0,1]} |f|
d\lambda < \infty$ gelten (absolute Integrierbarkeit).
Betrachten wir das Integral des Betrags:
$$ \int_{(0,1]} |f| d\lambda = \sum_{n=1}^{\infty} \int_{I_n}
\left| (-1)^n \frac{1}{x} \right| dx = \sum_{n=1}^{\infty}
\int_{1/(n+1)}^{1/n} \frac{1}{x} dx $$
Wie oben berechnet, ist dies:
$$ \sum_{n=1}^{\infty} \ln\left( 1 + \frac{1}{n} \right). $$
Wir nutzen die Abschätzung $\ln(1+x) \approx x$ für kleine $x$.
Genauer gilt für $x \in (0, 1]$ die Ungleichung $\frac{x}{2} \le
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\ln(1+x) \le x$.
Hier ist $x = \frac{1}{n}$. Also verhält sich die Reihe wie die
harmonische Reihe:
$$ \sum_{n=1}^{\infty} \ln\left( 1 + \frac{1}{n} \right) \geq
\sum_{n=1}^{\infty} \frac{1}{2n}. $$
Die harmonische Reihe divergiert gegen $+\infty$.
Daher ist $\int |f| d\lambda = \infty$, und $f$ ist nicht
Lebesgue-integrierbar.

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\subsection*{Hausaufgabe 5 (13 Punkte)}
\textbf{Aufgabe:}
Konstruieren Sie einfache Funktionen $f_n, g_n: (0,1) \rightarrow
\mathbb{R}$ mit $f_n(x) \leq \frac{1}{\sqrt{x}} \leq g_n(x)$ und
zeigen Sie, dass die Grenzwerte der Integrale übereinstimmen.
Bestimmen Sie $\int_{(0,1)} \frac{1}{\sqrt{x}} d\lambda$.
Hinweis: $(0,1) = \bigcup_{k \in \mathbb{N}} J_k$ mit $J_k = [(1-
\epsilon)^{2k}, (1-\epsilon)^{2(k-1)})$ für $0 < \epsilon < 1$.

\textbf{Lösung:}
Wir folgen dem Hinweis. Sei $\epsilon \in (0,1)$ fest.
Setze $q = (1-\epsilon)^2$. Dann ist $q \in (0,1)$.
Die Intervalle sind $J_k = [q^k, q^{k-1})$.
Die Länge von $J_k$ ist $\lambda(J_k) = q^{k-1} - q^k = q^{k-1}(1-
q)$.

Auf dem Intervall $J_k$ variiert $x$ zwischen $q^k$ und $q^{k-1}$.
Da $h(x) = \frac{1}{\sqrt{x}}$ monoton fallend ist, gilt für $x
\in J_k$:
$$ \frac{1}{\sqrt{q^{k-1}}} \leq \frac{1}{\sqrt{x}} \leq \frac{1}
{\sqrt{q^k}}. $$
Vereinfacht ($q = (1-\epsilon)^2 \implies \sqrt{q} = 1-\epsilon$):
$$ \frac{1}{(1-\epsilon)^{k-1}} \le \frac{1}{\sqrt{x}} \le
\frac{1}{(1-\epsilon)^k}. $$
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Wir definieren nun einfache Funktionen für ein festes $N$ (als
Index für den Grenzwertprozess der feineren Zerlegung oder wir
summieren über alle Intervalle, der Aufgabentext verlangt $f_n$
und $g_n$, interpretieren wir $n$ als Parameter für $\epsilon_n
\to 0$ oder wir definieren die Funktionen als abzählbare Summen,
wobei "einfache Funktion" meist endliche Summe bedeutet. Wir
betrachten hier die Reihe über alle $k$ als Limes von einfachen
Funktionen):

Sei $\phi_\epsilon = \sum_{k=1}^\infty \frac{1}{(1-\epsilon)^{k-
1}} \chi_{J_k}$ (Untersumme)
Sei $\psi_\epsilon = \sum_{k=1}^\infty \frac{1}{(1-\epsilon)^k}
\chi_{J_k}$ (Obersumme)
(Dies sind streng genommen Grenzwerte einfacher Funktionen, aber
für das Integral reicht die Betrachtung der Reihen).

Berechnen wir das Integral der Untersumme:
$$ I_{unten}(\epsilon) = \sum_{k=1}^\infty \frac{1}{(1-
\epsilon)^{k-1}} \lambda(J_k) = \sum_{k=1}^\infty \frac{1}{(1-
\epsilon)^{k-1}} (1-\epsilon)^{2(k-1)} (1 - (1-\epsilon)^2) $$
$$ = \sum_{k=1}^\infty (1-\epsilon)^{k-1} (2\epsilon - \epsilon^2)
= (2\epsilon - \epsilon^2) \sum_{j=0}^\infty (1-\epsilon)^j $$
Dies ist eine geometrische Reihe $\sum q^j = \frac{1}{1-q}$ mit
$q=1-\epsilon$:
$$ I_{unten}(\epsilon) = (2\epsilon - \epsilon^2) \cdot \frac{1}{1
- (1-\epsilon)} = \frac{2\epsilon - \epsilon^2}{\epsilon} = 2 -
\epsilon. $$

Berechnen wir das Integral der Obersumme:
$$ I_{oben}(\epsilon) = \sum_{k=1}^\infty \frac{1}{(1-\epsilon)^k}
\lambda(J_k) = \sum_{k=1}^\infty \frac{1}{(1-\epsilon)^k} (1-
\epsilon)^{2k-2} (2\epsilon - \epsilon^2) $$
$$ = (2\epsilon - \epsilon^2) \sum_{k=1}^\infty (1-\epsilon)^{k-2}
= (2\epsilon - \epsilon^2) \frac{1}{(1-\epsilon)}
```

```
\sum_{k=1}^\infty (1-\epsilon)^{k-1} $$
$$ = (2\epsilon - \epsilon^2) \frac{1}{1-\epsilon} \cdot \frac{1}
{\epsilon} = \frac{2 - \epsilon}{1-\epsilon}. $$

Nun konstruieren wir die Folge $(f_n)$ und $(g_n)$ durch Wahl
einer Folge $\epsilon_n \to 0$ (z.B. $\epsilon_n = 1/n$ für $n$
groß genug).
Dann gilt für die Integrale:
$$ \lim_{n \to \infty} \int f_n d\lambda = \lim_{\epsilon \to 0}
(2 - \epsilon) = 2. $$
$$ \lim_{n \to \infty} \int g_n d\lambda = \lim_{\epsilon \to 0}
\frac{2 - \epsilon}{1 - \epsilon} = 2. $$

Da $f_n \le \frac{1}{\sqrt{x}} \le g_n$ und die Integrale gegen
denselben Wert konvergieren, ist $\frac{1}{\sqrt{x}}$ Lebesgue-
integrierbar und der Wert des Integrals ist:
$$ \int_{(0,1)} \frac{1}{\sqrt{x}} d\lambda = 2. $$

\end{document}
```

> *CM2526_exercises_06 - PDF*

Solve all exercises of the loaded exercise sheet inlcuding recap of each actual task and a step by step derivation. The output should be in correct LaTeX syntax using the preamble given below. Put your output into a code block separated from your comments.

--- minimlal header for subfiles format--

\documentclass[../akolberesultsmaster.tex]{subfiles}

\begin{document}

> *Thinking:*
>
> > *Thinking:*
> >
> > ***Examining LaTeX Solutions***
> >
> > *I'm currently focused on the requirements for the "Exercise Sheet 6" solutions, specifically from the provided PDF. My priority is to generate accurate LaTeX code. I've begun to break down the problems from the document, prioritizing LaTeX formatting. I'm taking a close look at the PDF to extract each question and plan the best way to represent each.*
> >
> > ***Crafting LaTeX Output***
> >
> > *I'm now fully immersed in generating the LaTeX code, adhering to the specified preamble and code block format. I'm prioritizing clarity and precision in the technical content, aiming for a helpful tone tailored for physics experts. Each task is being addressed with step-by-step derivations*

and solutions, carefully incorporating image tags where necessary. The citation structure is also being implemented.

Detailing Specific Problems

I'm now fully immersed in the details of the remaining problems. I am generating LaTeX for task (c) and (d) from exercise 2, which requires sketching a Debye-Scherrer pattern and calculating relative ring diameters for simple cubic crystals. I'm carefully analyzing the data in exercise 3, which involves identifying crystal structures from given angles, and I'm preparing to calculate lattice constants. Exercise 4 focuses on LEED patterns. I'm focused on creating LaTeX to analyze the provided diffraction spots and calculate lattice constants for each material.

Analyzing Structure Factors

I've been working on Exercise 1, analyzing the structure factors for CsCl and CsI. I'm focusing on deriving the structure factor formula and applying it to calculate whether the (100) reflection is visible. I've considered the form factors for each atom and how they relate to the atomic numbers, which helps determine the scattering amplitude. This will help determine the presence or absence of specific reflections.

Determining Diffraction Patterns

I'm now calculating the ratios of the ring diameters for simple cubic crystals based on the Bragg condition and the relationship between scattering angle and the square root of the sum of squared Miller indices. I'm focusing on deriving the proportionality of the diameter with the square root of the index sum. I am going to apply these calculations for the exercise.

Matching Diffraction Patterns

I'm now analyzing the angle data from the provided samples to identify their crystal structures. I'm focusing on the ratios of $\sin^2(\theta)$ values to match known patterns for BCC, FCC, and diamond structures. Currently, I'm examining "Sample A" using the formula $N = h^2 + k^2 + l^2$ *, and comparing the calculated ratios to the standard values for various lattice structures to* *determine the best fit. I'm taking the ratios of $1 : 1.34 : 2.67 : 3.68$ and determining the pattern that best fits.*

Interpreting Diffraction Data

I've been analyzing the angle data from the provided samples to identify crystal structures. Focusing on Samples B and C, I'm now using the formula $N = h^2 + k^2 + l^2$, and comparing the calculated ratios of $\sin^2(\theta)$ values to known patterns for BCC, FCC, and diamond structures to determine the best fit. I've successfully identified Sample B as BCC, matching the sequence $1 : 2 : 3 : 4$ to the indices. Sample C strongly indicates a Diamond structure.

Comparing Lattice Constants

I'm now focusing on calculating the lattice constant ratios using the provided LEED pattern. I've noted the arrangement of spots: green (Ir) forming a middle ring, blue (Cs) an inner ring, and black (graphene) an outer ring. I'm focusing on their relative distances from the center, which are inversely proportional to their lattice constants. I plan to use this relationship to derive the lattice constant ratios.

Interpreting Reciprocal Lattice

I'm now focusing on deriving the relationships between lattice constants using the LEED pattern, considering the spot arrangement. I'm focusing on the green (Ir), blue (Cs), and black (graphene) rings. I'm noting the blue spots' positions relative to green and black along radial lines. I'm trying to determine if blue forms a superstructure and analyzing if the blue spots are at $1/3$ or $1/\sqrt{3}$ the distance from the origin. The question is whether they represent a $\sqrt{3} \times \sqrt{3}R30^\circ$ structure.

Refining Superlattice Analysis

I'm now carefully examining the LEED pattern, focusing on the blue spots' positions relative to the green (Ir) and black (graphene) rings. I've noted the aligned spots and considered the real-space interpretation: the blue spots seem to be at 1/3 the distance from the center as the green spots. I'm focusing on confirming a 3×3 real-space superlattice, implying $a_{Cs} = 3a_{Ir}$

and $k_{Cs} = k_{Ir}/3$. I'm carefully measuring distances and considering if the question hints at a specific pattern interpretation.

Determining Superlattice Relationships

I'm now back to examining the LEED pattern, and carefully re-evaluating the relationships between the blue, green, and black spots. I'm focusing on their radial distances from the center and the possibility of a superlattice structure. The blue (Cs) spots seem to be closer to the center, leading me to estimate the relationship between their reciprocal lattice vectors. I'm comparing the spots and re-assessing whether blue's distance is 1/3, 1/2, or another factor of green's. The image on page 2 provides helpful visual guidance.

Revising Superlattice Ratios

I'm now revising my analysis of the LEED pattern, guided by the provided hint and the image's layout. I've ruled out the $\sqrt{3} \times \sqrt{3}R30^\circ$ superstructure due to spot alignment. I'm focusing on the 3×3 option, which implies a direct relationship between the lattice constants of Cesium (Cs) and Iridium (Ir). I am verifying if the image supports $a_{Cs} = 3a_{Ir}$ or a commensurate relationship through visual radial distance verification of blue and green circles. I'm now looking at the ratio closely.

Analyzing Lattice Ratios

I'm now determining the ratios of lattice constants, focusing on the moiré pattern for Graphene on Ir(111). The problem implies a calculation, so I'm measuring spot distances to derive a_{Ir} . Based on the moiré pattern, $10a_G \approx 9a_{Ir}$. With $a_G = 2.46Å$, I calculate a_{Ir} at roughly 2.73 $Å$, in line with literature values. This standard Gr/Ir(111) setup seems to be assumed.

Determining Lattice Constant Ratio

I'm now focusing on deriving the real-space relationship between Cs and Ir lattice constants based on the LEED pattern. Considering the 3×3 moiré pattern, and the given values for graphene, I'm now aiming to determine the specific numerical relation for a_{Cs}/a_{Ir} . The moiré pattern, with its regular spacing, offers key clues for establishing the real-space ratio.

> > ***Determining Crystal Structures***
> >
> > *I'm now calculating lattice constants for Sample A, identifying it as FCC based on the matching sin²(θ) ratios. I've used Bragg's law and the known wavelength to get $a = 3.71 \mathring{A}$. This calculation is now validated and ready to move to other samples.*
> >
> > ***Finalizing LEED Analysis***
> >
> > *I'm now fully immersed in the final calculations for the lattice constants of Sample A, Sample B, and Sample C, confirming their structures as FCC, BCC, and Diamond, respectively. Using Bragg's law and the provided wavelength, I have precisely determined the lattice constants for each sample: FCC, with a lattice constant of roughly 3.71 $\mathring{A}$; BCC with approximately 4.38 $\mathring{A}$; and Diamond around 3.66 $\mathring{A}$. This concludes my detailed analysis.*
> >
> > ***Completing the Calculations***
> >
> > *I'm now fully immersed in the final calculations. I've double-checked my work for all sample calculations and derived equations. I've ensured a consistent approach for Sample A, B, and C, including using proper citations and clear derivations. I'm focusing on the overall readability. I'm writing out detailed calculations for the lattice constants of Sample A, Sample B, and Sample C, and writing the final summary.*
> >
> > ***Refining Lattice Constant Derivation***
> >
> > *I'm now focusing on deriving the specific relationship for the lattice constant of Cs in the Cs/Ir system, using the LEED image. Considering the reciprocal space, the positions of spots seem to imply that the blue spots are at 1/3 the radius of the green spots. I am also checking against the previous calculations.*

Here are the solutions to Exercise Sheet 6 for Condensed Matter Physics I.

```
\documentclass[../akolberesultsmaster.tex]{subfiles}
\usepackage{amsmath}
```

```
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage{booktabs}

\begin{document}

\section*{Exercise 1: Structure factors of CsCl and CsI}

\subsection*{Task Recap}
[cite_start]We are given that CsCl and CsI both crystallize in the
CsCl structure (simple cubic Bravais lattice with a two-atom
basis)[cite: 9]. [cite_start]Experimental X-ray diffraction shows
the (100) reflection is clearly visible for CsCl but clearly
absent (cannot be detected) for CsI[cite: 10]. [cite_start]We must
explain this result[cite: 11].

\subsection*{Solution}

The intensity of a Bragg reflection $(hkl)$ is proportional to the
square modulus of the structure factor $S_{\mathbf{K}}$, defined
as:
$$ S_{\mathbf{K}} = \sum_{j=1}^{N_{basis}} f_j(\mathbf{K}) e^{-i
\mathbf{K} \cdot \mathbf{d}_j} $$
where $f_j$ is the atomic form factor of atom $j$, and
$\mathbf{d}_j$ is its position in the unit cell.

**1. The CsCl Structure:**
The CsCl structure consists of a simple cubic lattice with a basis
of two atoms:
\begin{itemize}
    \item Atom 1 (e.g., Cs) at position $\mathbf{d}_1 = (0, 0,
0)$.
    \item Atom 2 (e.g., Cl or I) at position $\mathbf{d}_2 =
(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.
\end{itemize}
```

```
The reciprocal lattice vector for the reflection $(hkl)$ is
$\mathbf{K} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$. The
scalar product is $\mathbf{K} \cdot \mathbf{d}_2 = \pi(h+k+l)$.
The structure factor becomes:
$$ S_{(hkl)} = f_{Cs} e^0 + f_{X} e^{-i\pi(h+k+l)} = f_{Cs} +
f_{X} (-1)^{h+k+l} $$
where $X$ is either Cl or I.

**2. The (100) Reflection:**
For the $(100)$ reflection, $h=1, k=0, l=0$, so $h+k+l = 1$ (odd).
$$ S_{(100)} = f_{Cs} - f_{X} $$

**3. Atomic Form Factors:**
The atomic form factor $f$ scales roughly with the number of
electrons (atomic number $Z$) of the atom, especially at small
scattering vectors ($f(0) = Z$).
\begin{itemize}
    \item **For CsCl:**
    Cesium ($Z=55$) and Chlorine ($Z=17$).
    Since $Z_{Cs} \gg Z_{Cl}$, the form factors are significantly
different: $f_{Cs} \gg f_{Cl}$.
    $$ S_{(100)}^{CsCl} = f_{Cs} - f_{Cl} \neq 0 $$
    Therefore, the intensity $I \propto |S|^2$ is non-zero, and
the reflection is **visible**.

    \item **For CsI:**
    Cesium ($Z=55$) and Iodine ($Z=53$).
    These atoms are isoelectronic (ions $Cs^+$ and $I^-$ both have
54 electrons) and have very similar atomic numbers. Thus, their
scattering power is nearly identical:
    $$ f_{Cs} \approx f_{I} $$
    $$ S_{(100)}^{CsI} = f_{Cs} - f_{I} \approx 0 $$
    This results in **extinction** (destructive interference) for
the (100) reflection, making it undetectable.
```

```
\end{itemize}

---

\section*{Exercise 2: Laue and Debye-Scherrer methods}

\subsection*{Task Recap}
\begin{itemize}
    [cite_start]\item \textbf{(a)} Identify potential crystal structures from Laue images (a) and (b) and state what else is needed to determine the Bravais lattice and basis[cite: 18, 19].
    [cite_start]\item \textbf{(b)} Compare the X-ray spectrum required for Laue vs. Debye-Scherrer experiments[cite: 20].
    [cite_start]\item \textbf{(c)} Sketch the Debye-Scherrer pattern on a perpendicular area detector[cite: 21].
    [cite_start]\item \textbf{(d)} Calculate relative ring diameters for a simple cubic powder and explain their lattice plane origins[cite: 26].
\end{itemize}

\subsection*{Solution}

\textbf{(a) Laue Pattern Analysis:}
The Laue method reveals the symmetry of the crystal axis parallel to the beam.
\begin{itemize}
    \item **Image (a):** Shows 4-fold rotational symmetry. This indicates the beam is incident along a 4-fold axis. Possible structures include **Tetragonal** or **Cubic** (viewed along the [001] direction).
    \item **Image (b):** Shows 6-fold (or 3-fold) rotational symmetry. This indicates the beam is incident along a 3-fold or 6-fold axis. Possible structures include **Hexagonal**, **Trigonal**, or **Cubic** (viewed along the [111] body diagonal).
\end{itemize}
```

```
\textbf{Conclusion limits:} Definite conclusions about the specific Bravais lattice (e.g., fcc vs bcc) or specific basis are \textbf{not possible} from symmetry alone.
\textbf{Additional quantity needed:} To determine the full structure (lattice type and basis), one needs to analyze the **intensities** of the spots (to resolve the structure factor) and know the **wavelength range** of the incident beam to determine lattice constants (since Laue uses polychromatic light, $d$-spacing cannot be calculated directly from spot position without knowing which $\lambda$ caused it).

\textbf{(b) X-ray Spectra:}
\begin{itemize}
    \item **Laue Experiment:** Requires "white" X-ray radiation (a continuous **polychromatic** spectrum, e.g., Bremsstrahlung). This ensures that for various lattice planes with spacing $d$, there is a matching $\lambda$ in the spectrum satisfying Bragg's law $2d\sin\theta = n\lambda$ for fixed angles.
    \item **Debye-Scherrer:** Requires **monochromatic** X-ray radiation (a single wavelength, e.g., $K_\alpha$ line). Since the sample is a powder (all orientations present), we vary the orientation $\theta$ to find matches for the fixed $\lambda$.
\end{itemize}

\textbf{(c) Sketch of Debye-Scherrer Pattern:}
The pattern on a flat area detector perpendicular to the beam consists of **concentric circles (rings)** centered on the transmitted beam spot.

\textbf{(d) Simple Cubic (SC) Ring Diameters:}
For a simple cubic lattice, diffraction occurs for all integer indices $(hkl)$. The selection rule is simply $h^2 + k^2 + l^2 = N$.
From Bragg's law $2d \sin \theta = \lambda$ and $d = \frac{a}{\sqrt{N}}$, we have:
```

```
$$ \sin \theta = \frac{\lambda}{2a} \sqrt{h^2+k^2+l^2} = \frac{\lambda}{2a} \sqrt{N} $$
For small angles, the diameter $D$ of the ring on the detector is proportional to $2\theta \approx 2\sin\theta$.
$$ D \propto \sqrt{N} $$
The sequence of allowed $N$ for simple cubic is: 1, 2, 3, 4, 5, 6, 8 (7 is forbidden as sum of 3 squares).

The first four rings correspond to:
\begin{enumerate}
    \item $N=1$: Planes $\{100\}$. Diameter $\propto \sqrt{1} = 1$.
    \item $N=2$: Planes $\{110\}$. Diameter $\propto \sqrt{2} \approx 1.41$.
    \item $N=3$: Planes $\{111\}$. Diameter $\propto \sqrt{3} \approx 1.73$.
    \item $N=4$: Planes $\{200\}$. Diameter $\propto \sqrt{4} = 2$.
\end{enumerate}
**Relative Diameters:** $1 : \sqrt{2} : \sqrt{3} : 2$.

---

\section*{Exercise 3: A Debye-Scherrer Experiment}

\subsection*{Task Recap}
We have diffraction angles for three samples (A, B, C) known to be fcc, bcc, and diamond. [cite_start]We must identify which is which [cite: 33, 36] [cite_start]and calculate the lattice constant $a$ assuming $\lambda = 1.541 \mathrm{\AA}$[cite: 38, 39].

\subsection*{Solution}

We analyze the squared sine of the diffraction angle, $\sin^2(\theta)$, because Bragg's law implies:
```

```
$$ \sin^2(\theta) = \left( \frac{\lambda}{2a} \right)^2
(h^2+k^2+l^2) = C \cdot N $$
Assuming the table values are $2\theta$ (standard for diffraction
data), we calculate $\sin^2(2\theta/2) = \sin^2(\theta)$. We look
for ratios of integers allowed by the selection rules.

**Selection Rules:**
\begin{itemize}
    \item **BCC:** $h+k+l$ must be even. Allowed $N$: 2, 4, 6, 8,
10, 12... (Ratios: 1:2:3:4:5:6)
    \item **FCC:** $h,k,l$ all even or all odd. Allowed $N$: 3, 4,
8, 11, 12... (Ratios: 3:4:8:11)
    \item **Diamond:** FCC rules + basis factor. $h+k+l=4n$ or all
odd. Allowed $N$: 3, 8, 11, 16... (Ratios: 3:8:11:16)
\end{itemize}

\textbf{Analysis of Sample A:}
Angles ($2\theta$): $42.2^\circ, 49.2^\circ, 72.0^\circ,
87.3^\circ$.
$\theta$: $21.1^\circ, 24.6^\circ, 36.0^\circ, 43.65^\circ$.
$\sin^2(\theta)$:
1. $\sin^2(21.1^\circ) \approx 0.1296$
2. $\sin^2(24.6^\circ) \approx 0.1733$
3. $\sin^2(36.0^\circ) \approx 0.3455$
4. $\sin^2(43.65^\circ) \approx 0.4760$

Normalize by the first value ($0.1296$):
Ratio: $1 : 1.34 : 2.67 : 3.67$.
Multiply by 3 to check for FCC pattern: $3 : 4.01 : 8.00 : 11.00$.
This perfectly matches the FCC sequence ($3, 4, 8, 11$).
\textbf{Identification:} Sample A is \textbf{FCC}.

\textbf{Analysis of Sample B:}
Angles ($2\theta$): $28.8^\circ, 41.1^\circ, 50.8^\circ,
59.6^\circ$.
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$\theta$: $14.4^\circ, 20.55^\circ, 25.4^\circ, 29.8^\circ$.
$\sin^2(\theta)$:
1. $\sin^2(14.4^\circ) \approx 0.0618$
2. $\sin^2(20.55^\circ) \approx 0.1232$
3. $\sin^2(25.4^\circ) \approx 0.1840$
4. $\sin^2(29.8^\circ) \approx 0.2470$

Normalize by the first value ($0.0618$):
Ratio: $1 : 1.99 : 2.98 : 4.00$.
This matches the BCC sequence ($1, 2, 3, 4$).
\textbf{Identification:} Sample B is \textbf{BCC}.

\textbf{Analysis of Sample C:}
Angles ($2\theta$): $42.8^\circ, 73.2^\circ, 89.0^\circ,
115.0^\circ$.
$\theta$: $21.4^\circ, 36.6^\circ, 44.5^\circ, 57.5^\circ$.
$\sin^2(\theta)$:
1. $\sin^2(21.4^\circ) \approx 0.1331$
2. $\sin^2(36.6^\circ) \approx 0.3554$
3. $\sin^2(44.5^\circ) \approx 0.4919$
4. $\sin^2(57.5^\circ) \approx 0.7112$

Normalize by the first value ($0.1331$):
Ratio: $1 : 2.67 : 3.70 : 5.34$.
Multiply by 3 (assuming first is $N=3$): $3 : 8.01 : 11.09 :
16.03$.
This matches the Diamond sequence ($3, 8, 11, 16$).
\textbf{Identification:} Sample C is \textbf{Diamond}.

\textbf{(b) Lattice Constants:}
Using the first peak for each sample and $\lambda = 1.541
\mathrm{\AA}$:
$$ a = \frac{\lambda \sqrt{N}}{2 \sin \theta_{peak1}} $$

* **Sample A (FCC, N=3, $\theta=21.1^\circ$):**
```

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    $$ a_A = \frac{1.541 \sqrt{3}}{2 \sin(21.1^\circ)} \approx
\frac{2.669}{0.720} \approx \mathbf{3.71 \AA} $$
* **Sample B (BCC, N=2, $\theta=14.4^\circ$):**
    $$ a_B = \frac{1.541 \sqrt{2}}{2 \sin(14.4^\circ)} \approx
\frac{2.179}{0.497} \approx \mathbf{4.38 \AA} $$
* **Sample C (Diamond, N=3, $\theta=21.4^\circ$):**
    $$ a_C = \frac{1.541 \sqrt{3}}{2 \sin(21.4^\circ)} \approx
\frac{2.669}{0.730} \approx \mathbf{3.66 \AA} $$

---

\section*{Exercise 4: LEED pattern of graphene, Ir(111), and Cs}

\subsection*{Task Recap}
[cite_start]Based on the provided LEED pattern image showing spots
for Graphene (black), Ir(111) (green), and a Cs superstructure
(blue)[cite: 42, 43, 44]:
\begin{itemize}
    [cite_start]\item \textbf{(a)} Determine Bravais lattice and
relative orientation[cite: 51].
    [cite_start]\item \textbf{(b)} Determine $a_{Ir}$ given $a_G =
2.46 \mathrm{\AA}$[cite: 52].
    [cite_start]\item \textbf{(c)} Determine $a_{Cs}$[cite: 53].
    [cite_start]\item \textbf{(d)} Determine if Cs is
$\sqrt{3}\times\sqrt{3}$ or $3\times3$ relative to Ir[cite: 55].
\end{itemize}

\subsection*{Solution}

\textbf{(a) Structure and Orientation:}
The diffraction spots for all three components (Graphene, Ir, Cs)
exhibit 6-fold rotational symmetry. This indicates that all three
materials form **hexagonal (triangular) Bravais lattices** in real
space.
Furthermore, the diffraction spots for all three lie along the
```

```
same radial lines (rays emerging from the center). Since
reciprocal lattice vectors are perpendicular to real space planes
(or generally aligned for hexagonal systems), this implies that
the unit cell vectors of all three lattices are **parallel**
(aligned) in real space. There is no relative rotation between the
lattices.

\textbf{(b) Lattice Constant of Ir(111):}
In LEED (reciprocal space), the distance $k$ of a diffraction spot
from the center is inversely proportional to the real-space
lattice constant $a$:
$$ k \propto \frac{1}{a} \implies \frac{k_{Ir}}{k_G} = \frac{a_G}
{a_{Ir}} $$
From the visual evidence in the provided LEED pattern (comparing
the Green circle radius to the Black circle radius), the Green
spots (Ir) are slightly closer to the center than the Black spots
(Graphene).
Estimating from the image: The Ir spots are at approximately 90\%
of the distance of the Graphene spots. (Standard literature values
for this system are $a_G = 2.46 \AA$ and $a_{Ir} \approx 2.71
\AA$).
Calculation based on typical visual ratio ($k_{Ir}/k_G \approx
0.9$):
$$ a_{Ir} = a_G \cdot \frac{k_G}{k_{Ir}} \approx 2.46 \AA \cdot
\frac{1}{0.9} \approx \mathbf{2.73 \AA} $$

\textbf{(c) Lattice Constant of Cs:}
We observe the Blue spots (Cs). These are located significantly
closer to the center, indicating a larger real-space lattice
constant (superstructure).
Visually comparing the radial distance of the Blue spot ($k_{Cs}$)
to the Green spot ($k_{Ir}$):
The Blue spot appears to be at $1/3$ of the distance of the Green
spot.
$$ k_{Cs} \approx \frac{1}{3} k_{Ir} $$
```

```
Therefore:
$$ a_{Cs} = 3 \cdot a_{Ir} \approx 3 \cdot 2.73 \AA = \mathbf{8.19
\AA} $$

\textbf{(d) Real-space Relationship:}
We are deciding between a $\sqrt{3}\times\sqrt{3}$ and a
$3\times3$ superstructure.
\begin{itemize}
    \item A $\sqrt{3}\times\sqrt{3} R30^\circ$ structure would
result in diffraction spots that are rotated by $30^\circ$
relative to the substrate spots in the LEED pattern.
    \item A $3\times3$ structure results in spots aligned with the
substrate spots (no rotation) at $1/3$ the reciprocal distance.
\end{itemize}
**Observation:** The Blue Cs spots are **aligned** with the Ir and
Graphene spots (no rotation is observed).
**Conclusion:** The Cs superstructure corresponds to a $\mathbf{3
\times 3}$ superstructure with respect to the Ir(111) surface.

\end{document}
```